Example

```
foremost  -t gif,pdf  -o /tmp/recovered –i image.dd
```

This example searches the image image.dd for GIF and PDF files. If found, the recovered files are stored in the directory */tmp/recovered*. To learn more, refer to:

- *http://www.theillien.com/Sys_Admin_v12/html/v12/i09/a1.htm*
- *http://sid.ethz.ch/debian/foremost/foremost-1.5.6/foremost.conf*

Scalpel

Scalpel is a complete re-write of Foremost version 0.69, employing some innovative techniques to enhance performance and decrease memory usage. Similar to Foremost, it reads a database of header and footer definitions, and extracts matching files from a set of block devices or image files—which include raw images, AFF images, RAM dumps or swap dumps, irrespective of the underlying file system. But unlike Foremost, where you can give file types on the command line itself, Scalpel requires you to edit the configuration file containing the database of headers and footers. By default, all headers and footers are enabled, so you need to comment out all except those that you want recovered. To learn more, refer to: *http://www.howtogeek.com/howto/15761/recover-data-like-a-forensics-expert-using-an-ubuntu-live-cd/*.

Magic Rescue

Magic Rescue is another tool to recover deleted files of one or more types, from a block device. To recover a particular type of file, it first scans the input block device in read mode for the magic bytes associated with that type. If found, it calls an external program to recover the corresponding file. This procedure of recovering a particular file type is called a recipe, and it is documented in what is called a recipe file. Magic Rescue comes with a number of recipes meant for recovering various common file types. Also, you can write a recipe file to recover a specific file type. When compared to Foremost, Magic Rescue is slightly more advanced and complex, but at the same time, if a recipe associated with a file type is comprehensive, then Magic Rescue can recover the associated files more effectively, compared to Foremost.

Usage

Usage is as *magicrescue <options> devices*, where *<devices>* is a comma-separated list of block devices to be searched, and *<options>* a list of options to be used. Some of the important options are:

- *-r <recipe>*: It specifies a recipe name, recipe file, or directory containing recipe(s) to extract one or more file types.
- *-b <block_size>*: This is to scan files that start at a multiple of the *<block_size>* argument. The option applies only to recipes following it.
- *-d <output_dir>*: This specifies the output directory to put recovered files. Again, do not use an output directory present on the same block device that you are scanning to recover files.

Example

```
magicrescue -r jpeg-jfif -r jpeg-exif -d /tmp/output /dev/sda1
```

In this example, the block device */dev/sda1* is scanned to recover JPEG files into the */tmp/output* directory. It is possible that one ends up recovering thousands of files. In such cases, the *magicsort* utility can be of help; it sorts the files present in a directory by invoking the *'file'* utility on each and every file in the directory. To learn more, refer to:

- *http://www.itu.dk/people/jobr/magicrescue/manpage.html#name*
- *http://www.linux.com/archive/feature/126525*
- *http://www.linux.com/archive/feature/126525*

PhotoRec

This is another popular file/data recovery utility. Unlike Foremost, Scalpel and Magic Rescue, all of which are command-based, this one is a terminal-based utility, and hence more user-friendly. Originally intended for graphics files, it has been extended to other file types as well. PhotoRec can today recover most common photo formats, audio file formats such as MP3, and document formats such as PDF, HTML, MS-Office, etc. Also, it can recover such file types from a variety of media, including hard disks, CD-ROMs, DVDs, USB drives, and flash memory cards. Like other data-recovery tools, PhotoRec is safe to use, as it will never attempt to write to the drive or the memory card from which files need to be recovered from.

Before starting PhotoRec, you should ensure that the required media is available on the system. If you have a memory card, insert it in the card reader, and so on.

To begin with, PhotoRec will first ask a user for source media selection, from a list of available media on the system. Next, users have to choose the type of partition table on the selected media. Following this, a list of partitions on the selected media will be presented. Also, a menu of actions is presented at the bottom of the terminal (see Figure 1):

- *Search:* Starts the recovery process on the selected partition.
- *Options:* Presents a variety of PhotoRec options for the recovery process.